

APPIAN INTERN INTERVIEW EXPERINCE

Appian (pronounced App-ian) Corporation is an American cloud computing and enterprise software company headquartered in McLean, Virginia, part of the Dulles Technology Corridor. The company sells a platform as a service (PaaS) for building enterprise software applications. It is focused on lowcode development, process mining, business process management, and case management markets in North America, Europe, the Middle East, and Southeast Asia. Appian offers a lowcode platform with a visual interface and pre-built development modules. As with other low-code platforms, it enables businesses to create apps and automations using little or no code.

ONLINE ASSESSMENT:

The online assessment was held in glider.ai virtually and we had to solve 3 questions in 1 hr.

Question 1:

You will be given a set of paths where each path consists of a set of strings representing a path to a particular page starting from the 'Main' (homepage) element. You need to construct a tree representing every path given as input. The root node will be the 'Main' element. Finally print it out as per the required format. (Breadcrumbs problem)

Sample input:

```
arr = [['Main', 'Dashboard', 'Settings'], ['Main', 'Profile', 'Uploads'], ['Main', 'Dashboard', 'Products'], ['Main', 'Profile', 'Remove'], ['Main', 'Manager', 'Work'], ['Main', 'Manager', 'Review']]
```

Expected Output:

Main

- Dashboard
 - Settings
 - Products
- Profile
 - Uploads
 - Remove
- Manager
 - Work
 - Review

Question 2:

Tickets are sold to a concert in a particular order. The number of tickets, the step size, and the offset will be given. Number of Tickets: This is the total count of tickets available for the concert. Step Size: This is the number of tickets to skip before selecting the next ticket. For example, if the step size is 3, every fourth ticket will be chosen. (It can also be a negative number.) Offset: This is the starting position from which the ticket selection begins. It can be thought of as the

initial position in the sequence of tickets where the process starts. The challenge is to generate a sequence of ticket numbers based on these parameters, reflecting the order in which tickets are sold.

It is a variant of the famous Josephus problem but with negative step.

Question 3:

You are provided with a list of group names and a list of people belonging to each group. Additionally, you are given an array of strings representing individuals standing in a line. The task is to reorder the array such that:

1. If a person from a group is in the line, all other subsequent members of the same group should move to stand next to each other in the order they appear.
2. Once all members of a group are adjacent, replace their names with the group name.

I was able to solve all 3 questions within 1hr. 22 people were selected for interview.

INTERVIEW:

Technical Round 1:

This was a pair programming kind of round that happened for all students from CEG, but it was skipped for some students from MIT including me.

Technical Round 2:

This was a mix of technical and behavioral. As I had mentioned aspiring web developer in my resume, I was asked about what happens when an URL is typed in the browser. Then to name some data structures and their real-life examples completely away from computers. Then I was asked what my favorite subject was in general and in computer science and why. I explained why math was my favorite and why theory of computation was my favorite. Then, my favorite programming language and why. I said Scala and I was asked to list out its characteristics and about functional programming. Then, I was presented with a list of problems

- identify the odd ball among eight balls, where seven balls were of the same weight, and one was either heavier or lighter. The challenge was to find the odd ball using a balance scale with just two weighings.
- Pseudocode to distribute a total value according to weights in an array
- Bubble sort and its complexity
- Paranthesis matching

Following that I was asked about OOPS concepts and if I had used them in any of my projects. And then I was asked to explain what was the favorite project I did and why I chose Appian. Finally, I was given an opportunity to ask questions in general or about Appian.

Behavioral Round:

This round was with two interviewers. I was asked some basic questions regarding my academics and personal background. Here too I was asked about favorite subject, programming language and why. Here it was more in detail with multiple follow-ups to my responses. I was then asked to compare Python and Scala. Then, I was asked what was my favorite project (Turing Machine simulator) out of those I put in my resume and the motivation behind it. Then, I was asked why I chose Appian. They remarked that I seemed more research-oriented and why I wanted to get into software development. Finally, I was given an opportunity to ask questions.

In the end, 5 of us (4 of us from CEG and 1 from MIT) got selected for the position of Software Engineer Intern.

Tips:

- Enhance your problem solving skills so that when faced with any problem in the wild, you can solve it without asking what species it is.
- Be sincere and honest. Don't express interest in a topic either when asked or in the resume because it would sound good. Mention what you really like so that you can speak more confidently.
- Stay true to your interests. If your projects are aligned with your interests, it would give the impression of genuine interest.
- In subjective questions like project related or favorite subjects, there is no right or wrong. Be confident.
- Listen to the pre-placement talk carefully and try to learn as much as possible about the company. This can help you answer why you chose the company.
- Prepare questions for the interviewers if possible.

All the very best,

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